

The Impact of Hospital-Sponsored Health Plans on Readmission Rates

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PRELIMINARY

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March 7, 2026

Abstract

Hospital systems are increasingly vertically integrated (VI) with healthcare financing by sponsoring their own health plans. This paper studies whether admission to a (VI) hospital is associated with different 30-day unplanned readmission rate compared to standard non-VI hospitals. To address patient selection into hospitals, we develop an instrumental-variables design that leverages quasi-experimental variation in hospital assignment generated by (i) the ambulance transportation in emergency care episodes and (ii) the patient's relative distance to VI versus non-VI hospitals. We implement this identification strategy using administrative Medicare Advantage (MA) enrollment and claims data. We find that at the VI hospitals have lower risk-adjusted readmission rates than other hospitals, but this association attenuates as the level of hospital markets concentration rises.

Keywords: Health Economics, Medicare Advantage, Vertical Integration, Provider-Sponsored Health Plan, Hospital Readmission.

1 Introduction

Provider-sponsored health plans (PSHPs)—insurance plans sponsored by health systems that also deliver care—have become an increasingly important form of vertical integration in Medicare Advantage (MA). Proponents argue that tighter integration between insurer and provider can improve care coordination and reduce waste, while critics emphasize the potential for market power and limited competitive discipline. Despite growing policy interest, there remains limited evidence on whether hospital-sponsored plans translate into meaningfully different patient outcomes at the hospital level, and whether any differences depend on the competitive environment in which hospitals operate.

This paper studies one specific and policy-relevant dimension of quality: 30-day unplanned hospital readmissions following emergency admissions. Readmissions are costly for MA plans and Medicare, reflect both hospital care and discharge management, and are a central target of quality measurement and payment reforms. Using administrative data on emergency episodes, we document a robust descriptive pattern: patients treated at plan-sponsored hospitals have lower risk-adjusted readmission rates than patients treated at other hospitals, but the magnitude of this difference is smaller in more concentrated local hospital markets. Section ?? summarizes this descriptive evidence, and Section ?? presents an illustrative model that rationalizes these patterns through the interaction of internalized readmission costs and market concentration.

A key challenge is that descriptive comparisons need not have a causal interpretation. Even with rich clinical risk adjustment and beneficiary controls, unobserved patient severity, referral patterns, and plan- and provider-level selection may jointly influence both the destination hospital (plan-sponsored versus non-sponsored) and the probability of readmission. To address this identification problem, we use two quasi-experimental instruments commonly employed in the emergency care and hospital choice literature: (i) an ambulance-based instrument that exploits quasi-random assignment to ambulance providers with different propensities to transport patients to plan-sponsored hospitals; and (ii) a differential-distance instrument that captures plausibly exogenous variation in proximity to plan-sponsored versus non-sponsored hospitals. These instruments shift admission to plan-sponsored hospitals while holding fixed observed patient risk and local market conditions, allowing us to estimate causal effects for the subset of emergency patients whose destination hospital is influenced by emergency transport practices and/or proximity.

Why readmissions and market structure are central. Readmissions are both clinically meaningful and economically consequential. They are a major source of potentially

avoidable utilization, and even modest reductions in readmission rates can generate substantial savings given the scale of Medicare inpatient spending. Moreover, Medicare explicitly targets readmissions through the Hospital Readmissions Reduction Program (HRRP), which links hospital reimbursement to risk-adjusted readmission performance and can reduce Medicare inpatient payments by up to a statutory cap. At the same time, the competitive environment in which hospitals operate is widely viewed as a key determinant of cost containment and efficiency: competition can discipline prices and may improve performance incentives, whereas high concentration can weaken those pressures. These considerations motivate a focus on how vertical integration interacts with local market structure in shaping readmission outcomes.

Mechanism and intuition. The illustrative model in Section ?? provides a simple mechanism consistent with the descriptive gradient. Plan sponsorship changes hospital incentives because the integrated entity internalizes the medical-cost consequences of readmissions for an additional mass of in-system episodes associated with the sponsor plan. This internalization increases the returns to investing in readmission-prevention effort (discharge planning, follow-up, care coordination), leading to lower readmission risk for admissions treated at plan-sponsored hospitals. The same model also predicts an attenuation with concentration: when local markets are highly concentrated and the sponsor hospital has high baseline market share, plan sponsorship induces less incremental “in-system exposure” for which readmission costs are internalized, so the incremental effort incentive from integration fades.

What the descriptive evidence shows. The motivating evidence in Section ?? documents two robust patterns in risk-adjusted readmission rates. First, at the average level of concentration, plan-sponsored hospitals have lower readmission rates than non-sponsored hospitals. Second, the difference attenuates as hospital markets become more concentrated (and is close to zero in highly concentrated markets). These patterns are consistent with the model’s predictions, but they are not causal: admission to a plan-sponsored hospital may be correlated with unobserved patient severity or other confounders that affect readmissions.

Empirical strategy and contribution. Our empirical framework is designed to move from descriptive patterns to causal evidence. We estimate instrumental-variables models that treat admission to a plan-sponsored hospital as endogenous and exploit quasi-experimental variation in hospital assignment generated by ambulance referral patterns and relative proximity. We combine this IV strategy with market fixed effects and rich patient and plan controls to address confounding from persistent market characteristics and plan-level differences

in care management. To characterize heterogeneity by market structure in a transparent way, we complement continuous interactions with a robustness analysis based on concentration bins (reported in the Appendix alongside first-stage and balance diagnostics).

The paper contributes to the literature on hospital–insurer vertical integration by providing a clinically and policy-relevant outcome (readmissions), a clear mechanism for why integration may improve readmission prevention, and an empirical design aimed at identifying causal effects and their dependence on local market concentration.

Roadmap. Section ?? describes MA and provider-sponsored plans. Section ?? presents descriptive evidence on readmissions by hospital sponsorship status and market concentration. Section ?? introduces an illustrative model. Section 5 describes the empirical framework and identification strategy. Section ?? reports causal estimates and heterogeneity by concentration. Section ?? concludes.

2 Institutional Setting

2.1 Medicare Advantage and provider-sponsored plans

Medicare Advantage (MA) is the alternative to traditional fee-for-service Medicare and covers its beneficiaries through contracts with private plans. Integration between the provider and the health plan is becoming a prominent feature of the Medicare Advantage (MA) program. As MA plans are paid on a capitated basis and receive additional bonus payments based on quality performance, there may be an increased incentive to provide integrated care across care settings in order to maintain healthcare cost below the capitated amount and improve their enrollees’ health to achieve the quality-based bonus.

Provider-sponsored health plans (PSHPs) are vertically integrated entities where a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. This organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals.

2.2 Outcome: 30-day unplanned hospital readmission

This paper focuses on hospital readmissions as a clinically meaningful and policy-relevant measure of care coordination and discharge quality. Readmissions are financially consequential for Medicare and hospitals, and they are explicitly targeted by Medicare’s Hospital

Readmissions Reduction Program (HRRP). Recent evidence indicates that a substantial share of acute admissions are followed by a rehospitalization within 30 days, with aggregate spending associated with readmissions on the order of tens of billions of dollars annually in Medicare.¹

Definition and construction. Our main outcome, RR_{it} , is an indicator equal to one if beneficiary i experiences an *unplanned* acute inpatient readmission within 30 days of discharge from the *index* emergency admission at time t , and zero otherwise. We treat multiple rehospitalizations within the 30-day window as a single event, consistent with HRRP-style targeted measures, which are most naturally interpreted as readmission probabilities rather than counts.² Following CMS-style measurement conventions, we exclude same-day transfers and implement a planned-readmission algorithm to remove rehospitalizations that are predictably scheduled (e.g., selected elective procedures) rather than reflecting failures of post-discharge management.³ Because our data are Medicare Advantage encounters, we can track readmissions to *any* acute-care hospital, not only the index hospital.

All specifications adjust for rich clinical risk at the patient level. In addition to demographics and a baseline risk score, we include Hierarchical Condition Category (HCC) comorbidity indicators and principal-diagnosis fixed effects for the index admission. This aligns with the spirit of CMS risk adjustment, which uses case-level models with extensive risk factors when computing risk-adjusted readmission measures.⁴

Why readmissions are informative for this setting. The policy motivation for using readmissions is that a non-trivial share of rehospitalizations may be avoidable through improved care transitions and coordination. In our context, this maps naturally to organizational features that affect discharge planning, follow-up management, and the integration of inpatient care with post-acute and outpatient settings.

Financial incentives and the HRRP. HRRP, implemented beginning in 2013, links hospital revenue to readmission performance. CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with “excess” readmissions relative to national benchmarks.⁵ The program initially targeted acute myocar-

¹See, e.g., ? for summary statistics and policy discussion.

²? notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

³? describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

⁴? describes CMS’s use of hierarchical logistic regression with a large set of patient risk factors for condition-specific readmission measures.

⁵? provides a detailed description of the HRRP penalty structure, including the use of risk-adjusted readmission rates over a three-year window and benchmarking against the national mean.

dial infarction, heart failure, and pneumonia, and later expanded to additional conditions and procedures.⁶ Penalties were phased in from a 1% cap in 2013 to 2% in 2014 and 3% from 2015 onward.⁷ Thus, readmissions affect hospital financial performance both directly (through costly utilization) and indirectly through explicit Medicare payment reductions tied to risk-adjusted readmission rates.⁸

3 Illustrative Model

This section develops a parsimonious illustrative model of hospital effort to prevent readmissions. The goal is not to fully characterize the contracting and pricing environment in Medicare Advantage, but to provide a transparent framework that (i) clarifies the incentives created by plan sponsorship and (ii) motivates the empirical objects studied in the next section. The model yields two testable qualitative predictions that guide the analysis:

- (i) admission to a vertically integrated (plan-sponsored) hospital is associated with greater readmission-prevention effort (and hence lower readmission risk) relative to admission to an otherwise similar non-integrated hospital;
- (ii) this VI advantage attenuates as local hospital markets become more concentrated, so that the difference in effort—and the implied difference in readmissions—shrinks toward zero in as the hospital-market concentration rises.

Consider a single index admission for a Medicare Advantage (MA) beneficiary treated at a hospital. After discharge, the beneficiary is readmitted with probability $p(e)$, which depends on the hospital's effort $e \geq 0$ (e.g., discharge planning, follow-up, care coordination). For transparency, assume a linear readmission technology

$$p(e) = \bar{p} - e, \quad e \in [0, \bar{p}], \quad \bar{p} \in (0, 1), \quad (1)$$

so $p'(e) = -1 < 0$. Effort is an *upfront* investment (organizational capabilities, staffing, protocols) with quadratic cost

$$C(e) = \frac{c}{2}e^2, \quad c > 0. \quad (2)$$

We also assume that VI and non-VI hospitals have the same technology of effort production.

⁶? summarizes the targeted conditions/procedures and the program timeline.

⁷?.

⁸? discusses how excess readmission rates translate into payment reductions under HRRP.

Baseline hospital profits under negotiated MA fees. Let $s_0 \in [0, 1]$ denote the hospital's baseline market share in the local hospital market (e.g., measured using beds or discharges). For patients enrolled in *other* MA plans (i.e., not the hospital's sponsor plan), the hospital is paid negotiated fees that may depend on local market conditions. Let $F^{NR}(s_0)$ and $F^R(s_0)$ denote the negotiated payments in the no-readmission and readmission states, respectively, and let w^{NR} and w^R be the corresponding treatment costs, with $w^R > w^{NR}$. Define per-episode profits

$$\pi^{NR}(s_0) \equiv F^{NR}(s_0) - w^{NR}, \quad \pi^R(s_0) \equiv F^R(s_0) - w^R, \quad (3)$$

and assume readmissions are privately undesirable for the hospital:

$$\Delta\pi(s_0) \equiv \pi^{NR}(s_0) - \pi^R(s_0) > 0. \quad (4)$$

A standard (non-integrated) hospital chooses e to maximize expected profits from its baseline volume, proportional to s_0 :

$$\Pi^S(e; s_0) = s_0 \left[(1 - p(e))\pi^{NR}(s_0) + p(e)\pi^R(s_0) \right] - \frac{c}{2}e^2. \quad (5)$$

Using $p'(e) = -1$, the first-order condition (FOC) is

$$\frac{\partial \Pi^S}{\partial e} = s_0 \left(\pi^{NR}(s_0) - \pi^R(s_0) \right) - ce = s_0 \Delta\pi(s_0) - ce = 0, \quad (6)$$

so the optimal effort for a standard hospital is

$$e^S(s_0) = \frac{s_0 \Delta\pi(s_0)}{c}. \quad (7)$$

Plan-sponsored (vertically integrated) hospital. Now consider a hospital that sponsors an MA plan. Let $\mu \in [0, 1]$ denote the share of MA beneficiaries in the local market who are enrolled in the sponsor plan. The sponsor plan receives a capitated payment b from Medicare for each beneficiary enrolled in its plan, *regardless of where enrollees are treated*. Thus, the hospital revenue of from its plan enrollees is μb .

The key incentive difference under vertical integration is that the integrated entity internalizes the medical cost consequences of readmissions for the subset of sponsor-plan enrollees whose care is delivered in-system and therefore affected by the hospital's effort. Let $(1 - s_0)h(\mu)$ denote the incremental mass of such "in-system" episodes induced by plan sponsorship, where $h(\cdot)$ is increasing and $h(0) = 0$. The factor $(1 - s_0)$ captures saturation: when

the hospital already has high baseline market share (high concentration / strong provider market power), sponsoring a plan adds less incremental exposure to the sponsor system's care-management technology.

Define the expected (resource) cost of an episode as

$$\tilde{w}(e) \equiv (1 - p(e))w^{NR} + p(e)w^R. \quad (8)$$

Using (1), we have

$$\tilde{w}'(e) = w^{NR} - w^R = -(w^R - w^{NR}) \equiv -\Delta w, \quad \Delta w > 0. \quad (9)$$

The plan-sponsored hospital's objective is

$$\Pi^{VI}(e; s_0, \mu) = s_0 \left[(1 - p(e))\pi^{NR}(s_0) + p(e)\pi^R(s_0) \right] + \mu b - (1 - s_0)h(\mu)\tilde{w}(e) - \frac{c}{2}e^2. \quad (10)$$

Differentiating and using (9) yields the FOC

$$\begin{aligned} \frac{\partial \Pi^{VI}}{\partial e} &= s_0 \Delta \pi(s_0) - (1 - s_0)h(\mu)\tilde{w}'(e) - ce \\ &= s_0 \Delta \pi(s_0) + (1 - s_0)h(\mu)\Delta w - ce = 0. \end{aligned} \quad (11)$$

Therefore, the optimal effort under vertical integration is

$$e^{VI}(s_0, \mu) = \frac{s_0 \Delta \pi(s_0) + (1 - s_0)h(\mu)\Delta w}{c}. \quad (12)$$

Model Predictions. Combining (7)–(12), the effort gap is

$$\Delta e(s_0, \mu) \equiv e^{VI} - e^S = \frac{(1 - s_0)h(\mu)\Delta w}{c}. \quad (13)$$

Since $h(\mu) \geq 0$ and $\Delta w > 0$, (13) implies:

1. **Vertical integration reduces readmissions.** For $\mu > 0$, $\Delta e(s_0, \mu) > 0$, so $e^{VI} > e^S$ and therefore $p(e^{VI}) < p(e^S)$ under (1). In words, plan sponsorship induces higher readmission-prevention effort because the integrated entity internalizes the medical-cost savings from avoiding readmissions for an additional mass of sponsor-plan episodes.
2. **The VI advantage attenuates with concentration.** Holding μ fixed, (13) implies $\Delta e(s_0, \mu) \rightarrow 0$ as $s_0 \rightarrow 1$. Intuitively, in highly concentrated markets the sponsor hospital is already a must-have provider (high baseline share), so plan sponsorship gen-

erates little additional in-system exposure for which the integrated entity internalizes readmission costs; as a result, the incremental effort incentive from vertical integration vanishes.

4 Descriptive Evidence

A central premise of the paper is that hospital–insurer vertical integration can reduce readmissions by strengthening incentives and capabilities for care coordination. In the illustrative model, greater internalization of the plan-side margin leads hospitals to choose higher care-coordination effort and therefore lower readmission risk. This section presents suggestive (non-causal) evidence consistent with that mechanism and highlights an additional empirical pattern: the association between vertical integration and readmissions is strongest in competitive markets and attenuates as local hospital markets become more concentrated.

4.1 Data and measurement

Provider-sponsored health plans and VI hospitals. To identify integrated MA plans sponsored by health systems, we use the AHRQ Compendium of Health Systems, which contains the list of all health plans offered by hospitals and health systems. We link the hospitals in the Compendium to their CMS provider identification numbers (CCN) in order to merge the VI classification to hospital-level datasets (including the AHA Annual Survey for hospital characteristics). To identify health plans sponsored by other provider organizations, we additionally search through the full list of MA contracts and flag those whose sponsor is a provider organization. Throughout the descriptive analysis, we classify a hospital-year as vertically integrated, $VIHosp_{ht} = 1$, if hospital h is affiliated with a provider-sponsored MA plan in year t .

Readmission outcomes and risk adjustment. We measure hospital readmission performance using risk-adjusted 30-day readmission rates from Medicare’s Hospital Readmissions Reduction Program (HRRP). Specifically, the dependent variable in the descriptive regressions is observed at the hospital $\times$ episode-type $\times$ year level, denoted RR_{het} , where episode types correspond to the HRRP condition categories. In the main specifications we condition on the standard HRRP risk-adjustment factors and additionally include a rich set of hospital and market controls described below.

Market concentration. To study how the VI–readmission association varies with local market structure, we compute the Herfindahl–Hirschman Index (HHI) using acute-care bed

shares. We consider two market definitions: Hospital Service Areas (HSA) in the main specification and counties as an alternative. Let m index the hospital’s market. We use $\ln(\text{HHI})_{mt}$ in the estimation and mean-center it to ease interpretation (so that the level effect of VI is evaluated at the average market concentration in the sample).

4.2 Reduced Form Results

We study how the within-hospital association between vertical integration and risk-adjusted readmissions varies with hospital market concentration. Let m index the hospital’s local market (HSA in the main specification). We estimate:

$$\text{RR}_{het} = \beta_1 \text{VIHosp}_{ht} + \beta_2 \ln(\text{HHI})_{mt} + \beta_3 \text{VIHosp}_{ht} \times \ln(\text{HHI})_{mt} + X'_{het}\gamma + \delta_e + \tau_t + \varepsilon_{het}, \quad (14)$$

where RR_{het} is the risk-adjusted 30-day readmission rate for hospital h , episode type e , and year t ; VIHosp_{ht} indicates whether hospital h is vertically integrated with a plan sponsor in year t ; and $\ln(\text{HHI})_{mt}$ is the (mean-centered) log HHI of the hospital’s market m in year t . The vector X_{het} includes a rich set of hospital and market controls (e.g., hospital size and ownership, teaching status, baseline case-mix, and local demographic and provider-supply measures). We include episode fixed effects δ_e and year fixed effects τ_t . Regressions are weighted by $\ln(\text{discharges})$ and standard errors are clustered at the hospital (CCN) level.

Table 1 summarizes the key coefficients. Two facts stand out. First, at the average level of concentration (i.e., when $\ln(\text{HHI})_{mt} = 0$), vertically integrated hospitals have lower readmission rates than otherwise similar non-integrated hospitals ($\beta_1 < 0$). Second, the interaction term is positive and statistically significant ($\beta_3 > 0$), implying that the VI–non-VI readmission gap becomes smaller as hospital markets become more concentrated.

Because $\ln(\text{HHI})$ is mean-centered, the coefficient on **VI hospital** captures the VI–non-VI difference evaluated at the **average** HHI in the estimation sample. In column (1) (HSA markets) the estimate -0.297 implies that VI hospitals have readmission rates that are 0.297 percentage points lower than non-VI hospitals at mean HHI; the analogous estimate in column (2) (county markets) is -0.288 percentage points. The positive and statistically significant interaction term, **VI hospital** $\times$ $\ln(\text{HHI})$, implies that this VI advantage *shrinks* as market concentration increases. Using the point estimates, a 10% increase in HHI relative to the mean reduces the magnitude of the VI effect by about $0.355 \times \ln(1.10) \approx 0.034$ in column (1) and $0.297 \times \ln(1.10) \approx 0.028$ in column (2). Interpreting these in percentage terms, this corresponds to an attenuation of roughly $100 \times (\exp(0.034) - 1) \approx 3.5\%$ (HSA) and $100 \times (\exp(0.028) - 1) \approx 2.9\%$ (county) in the VI–non-VI readmission gap.

Table 1: Readmission rates, vertical integration, and market concentration

	(1) Market = HSA	(2) Market = County
VI hospital	-0.297*** (0.105)	-0.288*** (0.104)
ln(HHI)	-0.135 (0.190)	-0.111 (0.178)
VI hospital $\times$ ln(HHI)	0.355*** (0.135)	0.297*** (0.114)
HS hospital	-0.365*** (0.099)	-0.378*** (0.099)
HS hospital $\times$ ln(HHI)	0.034 (0.140)	0.155 (0.105)
ln(Expected RR)	14.879*** (0.601)	15.045*** (0.599)
ln(Risk score)	2.636*** (0.460)	2.790*** (0.463)
Episode FE	Yes	Yes
Year FE	Yes	Yes
Weights	ln(discharges)	ln(discharges)
Clusters (CCN)	1,941	1,941
Observations	85,610	75,679

Notes: Unit of observation is hospital $\times$ episode type $\times$ year. The dependent variable is the risk-adjusted 30-day readmission rate. ln(HHI) is the log Herfindahl–Hirschman Index constructed from acute-care bed shares and mean-centered in the estimation sample. Markets are defined as HSAs in column (1) and counties in column (2). Both specifications include episode and year fixed effects and analytic weights equal to ln(discharges). A rich set of hospital and market-level controls are included but not reported. Standard errors (in parentheses) are clustered at the hospital (CCN) level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Taken together, these estimates imply the following descriptive pattern: VI hospitals are associated with lower risk-adjusted readmission rates than non-VI hospitals in competitive markets, but this association fades as market concentration rises and is close to zero in highly concentrated markets. We do not interpret these correlations as causal since hospital integration status is endogenous, and unobserved differences across integrated and non-integrated hospitals (or across markets) could drive the observed gradient. This motivates the instrumental-variables design in the next section, which isolates plausibly exogenous variation in admission to a VI hospital and (in robustness) variation in exposure to provider-sponsored plans.

Figure 1 provides a transparent visualization of this relationship. The figure residualizes readmission rates on the full set of controls and year/episode fixed effects, then plots binned

means against HSA-level HHI separately for VI and non-VI hospitals, along with within-group linear fits. The VI series starts below the non-VI series in competitive HSAs, but rises more steeply with HHI, so the vertical gap narrows in highly concentrated HSAs.

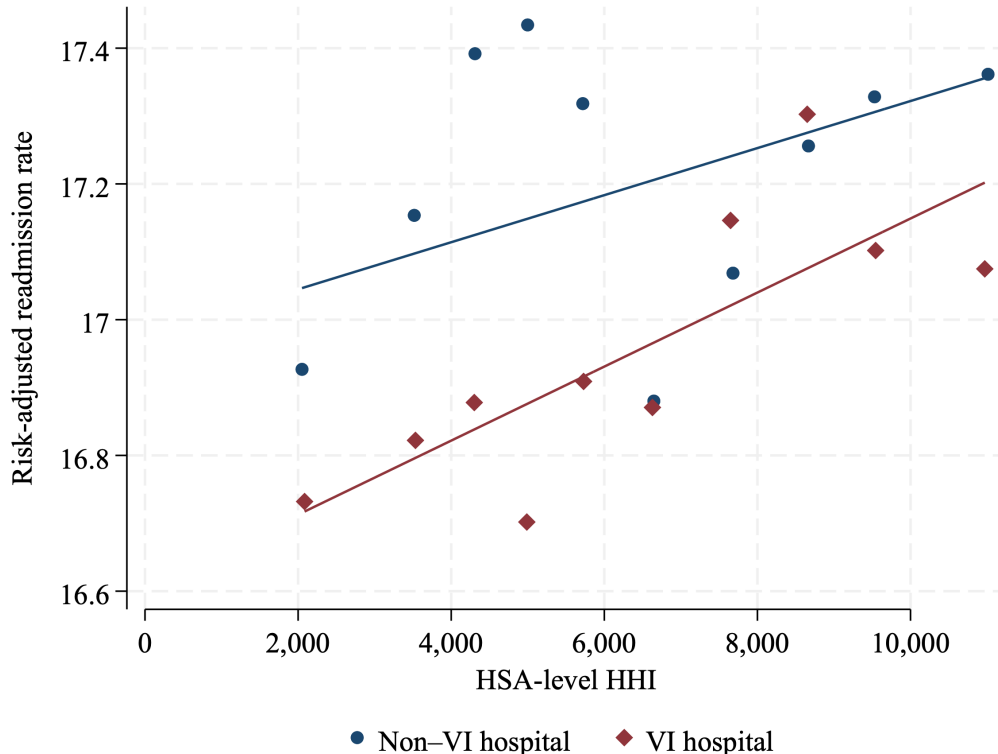


Figure 1: Readmission Rate vs. HHI by VI Hospital Status

Notes: Each point shows the mean readmission rate within each decile of the HSA-level HHI distribution, separately for VI and non-VI hospitals. The plotted outcome is residualized on a rich set of hospital and market controls, year fixed effects, and episode indicators, using $\ln(\text{discharges})$ as analytic weights. The HSA-level HHI is constructed from acute-care bed shares. Lines are within-group linear fits.

4.3 Patient Selection and Administrative Data

Even with a rich set of controls, a comparison of readmissions between plan-sponsored and non-sponsored hospitals can be confounded by selection. In emergency care settings, hospital assignment is influenced by ambulance routing protocols, proximity, and network considerations, but it may also respond to unobserved patient severity and clinical complexity not fully captured by claims-based risk adjustment. For example, patients with more complex discharge needs may be preferentially transported to particular hospitals; similarly, unobserved social risk and post-acute care access may affect both hospital choice and readmissions. These unobservables generate correlation between D_i and the structural error term

in outcome equations, biasing ordinary least squares (OLS) estimates. The empirical strategy therefore uses instrumental variables to generate quasi-random variation in admission to plan-sponsored hospitals.

Addressing patient selection requires individual-level information on beneficiary risk and utilization that is not available in hospital-level aggregates. In particular, even the descriptive patterns in Section ?? rely on constructing *risk-adjusted* readmission outcomes and conditioning on rich clinical and demographic controls at the episode level. We therefore use administrative Medicare Advantage enrollment and encounter/claims data to (i) observe plan enrollment at the time of admission, (ii) measure baseline risk and comorbidities used in the risk adjustment X_{het} , and (iii) identify the admitting hospital and subsequent readmissions. We then aggregate episodes to the hospital $\times$ episode-type $\times$ year level to form risk-adjusted readmission rates.

Medicare Advantage Claims Data We use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan. The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. I exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans. The main data source for patients’ acute-care utilization is the Inpatient Encounter Data. These files identify the admitting hospital (CCN) and, for ambulance-transported episodes, the ambulance provider. They include ICD-10 diagnosis and procedure codes, length of stay, and discharge disposition, which we use to construct episode definitions, risk adjusters, and readmission outcomes. In addition, we use the Inpatient, Outpatient, Skilled Nursing Facility (SNF), Home Health Agency (HHA), Durable Medical Equipment (DME), and Carrier Encounter files to measure broader utilization and to construct spending-based measures in supplementary analyses.

5 Empirical Framework

The unit of observation is an emergency care episode indexed by i , defined as an emergency admission (or ED-to-inpatient transfer) for an MA beneficiary in year $t(i)$. The primary outcome is an indicator for 30-day unplanned readmission,

$$y_i \in \{0, 1\}.$$

Let $h(i)$ denote the admitting hospital. The endogenous treatment is an indicator for whether the admitting hospital is plan-sponsored:

$$D_i \equiv \mathbb{1}\{h(i) \text{ is a plan-sponsored (PSHP) hospital}\}.$$

We condition on a rich vector of patient and episode controls X_i (e.g., demographics, risk score / HCCs, comorbidity flags, diagnosis or condition fixed effects, and time controls), and on plan controls (see below).

Markets are indexed by m , constructed at the county (or other pre-specified) level. Let $m(i)$ denote the market in which hospital $h(i)$ operates. We measure market concentration using a predetermined Herfindahl–Hirschman Index (HHI) constructed from hospital capacity or utilization shares in a pre-period:

$$M_m \equiv \ln(\text{HHI}_m),$$

where higher M_m corresponds to more concentrated markets. Because M_m is predetermined and varies only at the market level, we include market fixed effects α_m in all baseline specifications; these fixed effects absorb the level effect of M_m and other time-invariant market characteristics.

We estimate a linear probability model with market fixed effects and allow the causal effect of admission to a plan-sponsored hospital to vary with market concentration. The structural equation is

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 (D_i \times \tilde{M}_{m(i)}) + X_i' \gamma + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i)} + \varepsilon_i, \quad (15)$$

where $\tilde{M}_m \equiv M_m - \bar{M}$ is the mean-centered concentration index, τ_t are year fixed effects, and $\phi_{\text{plan}(i)}$ are plan fixed effects.⁹ Mean centering implies that β_1 is the causal effect evaluated

⁹In alternative specifications, we replace plan fixed effects with contract-by-year fixed effects to absorb time-varying differences in benefit design and utilization management at the contract level. In that case, we retain year fixed effects to capture common time shocks not absorbed by the contract-by-year controls.

at the average concentration level in the sample, while β_2 captures the slope of heterogeneity with respect to concentration. Because M_m is absorbed by market fixed effects, we do not interpret concentration causally; rather, $\tilde{M}_m$ indexes how the VI-hospital effect varies systematically across market structures.

Fixed effects and their role in identification. The fixed effects in (15) serve two purposes. First, market fixed effects α_m absorb all time-invariant differences across local hospital markets that could simultaneously affect (i) the probability of admission to a plan-sponsored hospital and (ii) baseline readmission risk, such as persistent differences in provider quality, practice style, post-acute care supply, local socioeconomic conditions, and the general level of hospital market concentration.¹⁰ Put differently, with α_m in place, the remaining identifying variation comes from differences in patients’ admitting hospitals *within the same market*, holding fixed all market-level confounders that do not vary over time.

Second, plan fixed effects $\phi_{\text{plan}(i)}$ absorb persistent differences across MA plans that could affect readmissions directly (e.g., utilization management intensity, care coordination programs, network design, and enrollee composition) and may also influence which hospitals enrollees are likely to use. Year fixed effects τ_t absorb common time shocks affecting readmissions and hospital practice (e.g., coding changes, broad shifts in clinical protocols, or policy changes affecting all markets). In specifications with contract-by-year fixed effects, we allow these plan-level differences to vary flexibly over time at the contract level, further tightening identification by absorbing contract-specific year-to-year changes in benefits or care management that could otherwise contaminate comparisons across hospital types.

Together, these fixed effects substantially reduce scope for confounding from time-invariant market characteristics and persistent plan features, so that the remaining endogeneity concern is primarily *patient selection into hospitals within market and within plan*. This motivates the instrumental-variables strategy described in Section ??.

Throughout, standard errors are clustered at a level appropriate for the assignment mechanism (e.g., at the ambulance provider level for the ambulance IV sample, and at the residential ZIP or market level for the distance IV), with robustness to alternative clustering choices.

¹⁰When concentration is measured using a predetermined market-level HHI that is constant (or nearly constant) over the study period, α_m also absorbs the level effect of $\ln(\text{HHI}_m)$, ensuring that identification of β_2 relies on within-market variation in D_i interacted with cross-market differences in $\tilde{M}_m$, rather than on levels comparisons across markets.

5.1 Endogeneity and instruments

The hospital level VI integration status could be endogenous for several reasons: (i) beneficiaries may select into VI hospitals based on unobserved preferences or health risk, and (ii) even in emergencies, unobserved severity and local provider attributes may correlate with being admitted to an integrated hospital. We therefore use an IV strategy with excluded instruments that generate quasi-experimental variations in the VI-hospital admission.

Ambulance-preference shifter for VI-hospital admission. The locus of treatment for emergency hospitalizations is, to a large extent, determined by pre-hospital factors, including ambulance transport decisions and patient location. Moreover, there is wide evidence that large city areas are often served by multiple ambulance companies, whereas smaller rural with a single ambulance company, neighboring companies provide service when the principal ambulance units are busy under so-called “mutual aid” agreements. Thus, in both cases, the assignment of the ambulance company to the patient is effectively random. (Doyle 2015, Chiang, et al., 2006 Ragone 2012).

In addition to plausibly exogenous assignment, ambulance companies are expected to have preferences for particular hospitals. In survey work described in Doyle et al. (2015), we found that paramedics have developed relationships with local emergency departments. For example, Skura (2001) studied ambulance assignment in the wake of a new system of competition between public and private ambulances in New York City. He found that patients living in the same ZIP code as public Health and Hospital Corporation hospitals were less than half as likely to be taken there when assigned a private, non-profit ambulance (29%) compared to when the dispatch system assigned them to an FDNY ambulance (64%). In most cases, the private ambulances were operated by non-profit hospitals and stationed near or even within those facilities, so they tended to take their patients to their affiliated hospitals. More broadly, with the exception of acute trauma care for which there are often defined local protocols for hospital assignment (Kahn et al. 2008) transport assignment for other emergencies is more likely to be driven by idiosyncratic preferences. Thus, in this context patient transport decisions are more likely to be made in ways that are not systematically tied to the underlying health of the patient.

Figure 2 illustrates the key source of quasi-random variation in hospital assignment that underlies the empirical strategy. When a medical emergency occurs, the dispatch system typically assigns a patient to the nearest available ambulance, which is effectively random from the patient’s perspective. Conditional on the patient’s location, the ambulance dispatched may have systematically different “preferred” hospitals due to historical routing patterns, contractual relationships, or local familiarity.

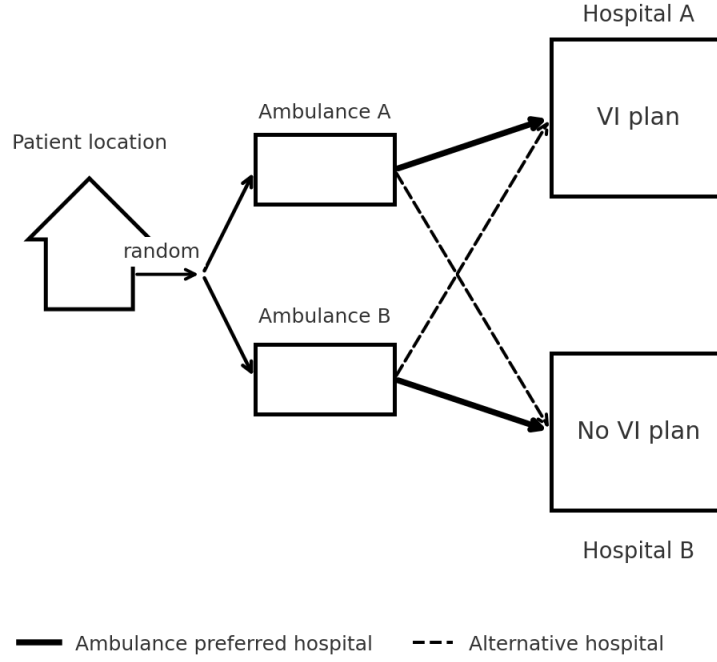


Figure 2: Hospital Assignment and Ambulance Preferences

As shown in Figure 2, two ambulances (A and B) serve the same geographic area but exhibit different propensities to transport patients to different hospitals. Ambulance A tends to deliver patients to Hospital A, which is vertically integrated (VI) with a Medicare Advantage plan, whereas Ambulance B tends to deliver patients to Hospital B, which is not integrated. Importantly, ambulances occasionally transport patients to the alternative hospital as well, generating natural variation in destination choice.

This setup creates a quasi-experimental shock in exposure to VI hospitals: patients who happen to be picked up by an ambulance with stronger ties to the VI hospital are more likely to be admitted there, even though the ambulance choice is conditionally random. This type of variation has been widely used in the hospital choice and emergency-care literature because it mimics an instrumental variable: ambulance identity predicts hospital assignment but is plausibly orthogonal to patient unobservables.

In other words, patients typically do not choose the ambulance, and dispatch/rotation protocols and ambulance company service areas generate quasi-random assignment to companies with heterogeneous hospital destination preferences. This yields plausibly exogenous variation in the admitting hospital conditional on residential location and origin, forming the basis for ambulance preference instruments (Doyle Jr et al., 2015).

Following the emergency-care literature, we use an ambulance-company “preference”

instrument that captures the propensity of ambulance company a to deliver patients to integrated hospitals. For each ambulance company a , market, year, and emergency condition, we compute the leave-one-out share of transports ending at an integrated hospital among other patients served by that company:

$$\text{AmbShare}_{it} = \frac{1}{N_{a(i),t,c(i)} - 1} \sum_{\substack{j:a(j)=a(i) \\ j \neq i}} \mathbf{1}\{\text{VI}_{jt}^h = 1\}, \quad (16)$$

where the summation is computed within the same market-year-emergency cell (to preserve relevance and comparability across episodes). Conditional on controls and fixed effects, we assume ambulance assignment affects outcomes only through the admitting hospital (and thus exposure to VI^h), not through unobserved severity or differential pre-hospital treatment.

Differential distance between VI and non-VI hospitals. We additionally use a geographic shifter based on relative proximity to integrated versus non-integrated hospitals. Let $\text{dist}(r(i), h)$ denote distance from the beneficiary’s residence (or pickup location) $r(i)$ to hospital h . Define the nearest plan-sponsored hospital $h^{VI}(i)$ and the nearest non-sponsored hospital $h^{NVI}(i)$, and construct

$$Z_i^{DD} \equiv \ln \text{dist}(r(i), h^{VI}(i)) - \ln \text{dist}(r(i), h^{NVI}(i)).$$

More negative values of Z_i^{DD} indicate greater relative proximity to a plan-sponsored hospital. The identifying intuition is that, in acute care settings, *distance is a first-order determinant of hospital assignment*: conditional on emergency condition, location, and market characteristics, patients (and ambulance routing protocols) tend to be transported to closer hospitals, and marginal differences in relative proximity shift the likelihood that the admitting hospital is plan-sponsored. In our setting, Z_i^{DD} therefore generates quasi-experimental variation in D_i by shifting the hospital set that is practically feasible for emergency admission, holding fixed rich patient risk adjustment and fixed effects.

Our exclusion restriction is that, conditional on the control set and fixed effects in (15), differential distance affects readmissions only through admission to a plan-sponsored hospital (and, in specifications that include plan controls, through plan enrollment channels already absorbed by those controls), not through unobserved severity or neighborhood-level confounders. The emergency/ambulance sample restrictions described below are designed to strengthen this restriction by limiting scope for discretionary hospital selection and for systematic differences in unobservables across beneficiaries living closer to VI versus non-VI hospitals.

Estimation samples and use of instruments. We construct the analysis sample in several steps. First, we restrict attention to non-discretionary emergency conditions—such as acute myocardial infarction, stroke, and hip fracture—that represent high-acuity clinical presentations for which hospital choice is effectively non-deferrable and largely outside the patient’s control. These conditions are clinically significant and account for a substantial share of inpatient utilization among Medicare beneficiaries.

Second, for the primary analysis we restrict the estimation sample to emergency hospitalizations arriving via ambulance transport. This restriction is central for the ambulance-based instrument (which relies on quasi-random assignment to ambulance providers with different destination propensities) and also strengthens the credibility of the differential-distance instrument by limiting scope for discretionary hospital selection by patients and families. Applying these restrictions yields a beneficiary-by-episode dataset. The unit of observation is an emergency episode for beneficiary i in calendar year t , defined as an inpatient admission originating in the emergency department (or an observation stay subsequently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, we observe the beneficiary’s MA plan enrollment at time t , the identity and characteristics of the admitting hospital, and utilization outcomes during and after the inpatient stay, including 30-day unplanned readmissions.

Our baseline IV estimates are obtained on the ambulance-transported emergency sample using the ambulance preference shifter as the primary instrument and, where informative, the differential-distance shifter as a complementary source of variation. Using both instruments on the same high-acuity ambulance sample keeps the identifying assumptions aligned across IV strategies and facilitates direct comparison of estimates.

Because differential distance does not mechanically require ambulance transport, we also report a robustness specification that expands the sample to *all* non-discretionary emergency admissions (including self-transport) and uses Z_i^{DD} as the excluded instrument. This broader emergency sample increases external validity and statistical power, while preserving the key feature that admissions are plausibly non-deferrable. In contrast, we do not use elective or discretionary (non-emergency) admissions as a primary estimation sample for the distance instrument, because hospital choice in such settings is more likely to reflect unobserved patient preferences, provider referrals, and plan network design, which can undermine the exclusion restriction for Z_i^{DD} . We therefore treat the emergency-based distance-IV estimates as the appropriate complement to the ambulance-IV design.

5.2 2SLS with heterogeneous effects

The presence of heterogeneity implies two endogenous regressors in (15): the indicator for admission to a plan-sponsored hospital D_i and its interaction with market concentration $D_i \times \tilde{M}_{m(i)}$. We therefore instrument the endogenous regressor vector

$$W_i \equiv \left(D_i D_i \times \tilde{M}_{m(i)} \right)$$

using the excluded instruments

$$Z_i \equiv \left(Z_i^A, Z_i^{DD}, Z_i^A \times \tilde{M}_{m(i)}, Z_i^{DD} \times \tilde{M}_{m(i)} \right),$$

along with the full set of included controls and fixed effects in (15).

What variation identifies each parameter. The coefficient β_1 is identified by instrument-induced shifts in admission to plan-sponsored hospitals that are orthogonal to unobserved patient severity, after conditioning on X_i and fixed effects. In practice, this variation comes from (i) quasi-random assignment to ambulance providers with different propensities to transport patients to plan-sponsored hospitals (captured by Z_i^A), and/or (ii) plausibly exogenous relative proximity to plan-sponsored versus non-sponsored hospitals (captured by Z_i^{DD}). Since $\tilde{M}_{m(i)}$ is predetermined and varies only at the market level (and is mean-centered), interacting instruments with $\tilde{M}_{m(i)}$ yields additional excluded variation that shifts the concentration-weighted treatment $D_i \times \tilde{M}_{m(i)}$. Intuitively, the same quasi-experimental shocks that move D_i generate larger movements in $D_i \times \tilde{M}_{m(i)}$ in more concentrated markets (higher $\tilde{M}_m$) and smaller movements in less concentrated markets (lower $\tilde{M}_m$). This interaction variation identifies β_2 , the slope of the treatment effect with respect to market concentration.

LATE interpretation. Under the standard IV assumptions (relevance, exclusion, and monotonicity), 2SLS identifies local average treatment effects (LATEs) for *compliers*: emergency patients whose destination hospital (VI vs. non-VI) is shifted by the instrument. With mean-centered $\tilde{M}_m$, the structural equation (15) can be written as

$$y_i = \beta_0 + (\beta_1 + \beta_2 \tilde{M}_{m(i)}) D_i + \dots,$$

so the implied causal effect of admission to a plan-sponsored hospital in market m is $\tau(m) \equiv \beta_1 + \beta_2 \tilde{M}_m$. The parameter β_1 should therefore be interpreted as the average complier effect evaluated at the mean concentration level in the sample (i.e., $\tilde{M}_m = 0$), while β_2 captures

how the complier effect varies systematically with market concentration. Equivalently, β_2 is the incremental change in the IV-identified treatment effect associated with a one-unit increase in $\tilde{M}_m$.

Instrument validation and robustness. First-stage strength, balance on predetermined characteristics, and robustness exercises—including nonparametric heterogeneity using concentration bins and alternative samples (e.g., restricting to non-PSHP enrollees)—are reported in Appendix ??.

6 Preliminary Results

Work in progress

References

Doyle Jr, J. J., Graves, J. A., Gruber, J., and Kleiner, S. A. (2015). Measuring returns to hospital care: Evidence from ambulance referral patterns. *Journal of Political Economy*, 123(1):170–214.

Appendix A

6.1 Instrument validity and robustness checks

This section reports two core diagnostics: instrument relevance (first stage strength) and balance on predetermined covariates. In addition, we implement a robustness check that estimates heterogeneity nonparametrically using concentration bins.

6.1.1 Instrument relevance

We report first-stage estimates for both endogenous regressors, D_i and $D_i \times \tilde{M}_{m(i)}$. For each instrument set (ambulance only, distance only, and the combined set), we present the first-stage coefficients and associated robust weak-instrument diagnostics, including the Kleibergen–Paap rk Wald F statistic. Because the model includes multiple endogenous regressors, we report first-stage strength for each equation as well as the joint strength of the excluded instruments for the endogenous regressor vector.

6.1.2 Balance on predetermined characteristics

To assess whether the instruments behave as-if random conditional on the included controls and fixed effects, we conduct balance tests on predetermined patient and episode characteristics. Specifically, we estimate auxiliary regressions of the form

$$X_{ik} = \eta_{0k} + \eta_{1k}Z_i^A + \eta_{2k}Z_i^{DD} + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i),t(i)} + \nu_{ik},$$

for key components of X_i (demographics, risk score / HCC summary, comorbidity indicators, and diagnosis/condition categories), and report standardized differences across instrument-induced variation. Under instrument validity, the excluded instruments should not systematically predict baseline risk once conditioning on location, time, plan-by-year fixed effects, and other controls. Balance is particularly relevant for the ambulance IV, where the identifying variation comes from ambulance providers with different destination propensities within the same pickup area and time period.

6.1.3 Robustness: heterogeneity by concentration bins

As a robustness check, we replace the continuous interaction in (15) with discrete concentration bins. Let $Q_m \in \{1, \dots, Q\}$ denote the market’s concentration bin (e.g., terciles) of

the predetermined M_m . We estimate

$$y_i = \beta_0 + \sum_{q=1}^Q \beta_q (D_i \times \mathbb{1}\{Q_{m(i)} = q\}) + X_i' \gamma + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i), t(i)} + \varepsilon_i, \quad (17)$$

instrumenting each interacted treatment $D_i \times \mathbb{1}\{Q_{m(i)} = q\}$ with the corresponding interacted instruments $Z_i^A \times \mathbb{1}\{Q_{m(i)} = q\}$ and $Z_i^{DD} \times \mathbb{1}\{Q_{m(i)} = q\}$ (and including the uninteracted instruments as needed for the level first stage). Equation (17) yields bin-specific IV effects that can be plotted against concentration bins to visualize the “fade-out” pattern without imposing linearity in M_m .

6.1.4 Plan controls and sample robustness.

Our baseline specifications include plan-by-year fixed effects to absorb plan-level utilization management and benefit design. As an additional robustness check, we re-estimate the IV models restricting the sample to beneficiaries enrolled in non-PSHP MA plans, thereby holding plan ownership constant and focusing on variation in the admitting hospital’s sponsorship status.